

Project Design Phase

Proposed Solution Template

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.N o.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Text-based learning interactions carry emotional signal - confusion, frustration, boredom, curiosity, confidence - that goes entirely uncaptured and unused. Students studying independently receive the same generic response whether they are confident or completely lost, and educators/TAs have no automated way to identify which students are struggling before they quietly disengage. This results in reduced engagement, delayed intervention, and students who don't get the kind of support that actually matches how they're feeling in the moment.
2.	Idea / Solution Description	An AI-driven emotion detection and personalized learning support platform (built with Streamlit) where a student describes their field of study and the learning challenge they're facing in plain text. Two independently trained models - a BiLSTM (TensorFlow/Keras) and a fine-tuned BERT (PyTorch + HuggingFace Transformers) - classify the underlying emotion across 5 classes (Bored, Confident, Confused, Curious, Frustrated), including detecting when more than one emotion is present at once. A keyword-based boosting layer sharpens ambiguous predictions. The detected emotion and confidence score are then passed to the Google Gemini API to generate a personalized, field-specific, empathetic response, with an automatic template-based fallback if the API is unavailable. Every interaction is logged, and a built-in analytics dashboard visualizes the student's emotional trend over the session.

S.No.	Parameter	Description
3.	Novelty / Uniqueness	• Runs two independently trained architectures (BiLSTM and BERT) side-by-side for direct comparison, rather than relying on a single model's judgment. • Detects and surfaces <i>mixed</i> emotions (e.g. "Bored + Frustrated") instead of forcing a single label, giving a more honest picture of how a student actually feels. • Purely text-based - no camera or microphone required - avoiding the added privacy sensitivity and hardware cost of facial or voice emotion detection. • Combines a rule-based keyword-boosting layer with two deep learning models, blending interpretable heuristics with learned predictions rather than relying on either approach alone.
4.	Social Impact / Customer Satisfaction	Helps students feel noticed and supported without needing to explicitly articulate "I need help," which can reduce quiet disengagement and dropout risk during independent study. Gives educators and teaching assistants a faster way to triage which students need attention, rather than manually reading and interpreting every message. Because it requires only text input, it remains accessible on low-end devices with no camera/microphone dependency, and does not collect or store any personally identifiable information, making it appropriate for a student audience.
5.	Business Model (Revenue Model)	As an academic/internship project, the current build has no active revenue model. A realistic future path could include: a freemium model for ed-tech platforms (basic single-model emotion detection free, dual-model comparison and analytics dashboard as a paid tier); B2B licensing as a plug-in for existing Learning Management Systems (e.g. Moodle, Canvas); or an API-as-a-service offering so other ed-tech developers can integrate emotion-aware responses into their own products.
6.	Scalability of the Solution	The modular codebase (separate preprocessing, BiLSTM, BERT, and keyword-boosting layers) allows any single component to be upgraded or replaced without rebuilding the rest of the system. The models could be retrained with additional emotion classes (e.g. a "Neutral" class for out-of-domain input) or extended to additional languages. Cloud deployment (e.g. Streamlit Community Cloud or Hugging Face Spaces) allows the application to scale to more concurrent users without infrastructure changes, and the same emotion-detection core could be exposed as an API for integration into other learning platforms at scale.